UHACK 4.0

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Problem statement ID : D01-S03

Problem statement : CROP HEALTH AND PEST MANAGEMENT

Theme : AGRICULTURE & AGRI TECH

PS Category : SOFTWARE

Team ID : UHK04217

Team Name : AGRIPLUS

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AGRIPLUS: *AI-Driven Early Pest Detection & Holistic Farm Management System*

• Proposed solution

An AI- Driven early pest detection and holistic farm management system is a smart agriculture solution that uses Artificial Intelligence (AI), Data, and digital tools to detect pest at an early stage and manage the entire farm efficiently - not just pests, but soil, crops , water, weather, and inputs as a whole.

- **Core Features**

  A Camera- based ML Module (MobileNetV2) that detects pests/ diseases from smartphone photos with highest accuracy .

- **Automate alerts**

  Generates instant " Red Alerts" for pest outbreaks (e.g.,Locusts/Fall Armyworm) and notifies nearby farmers via Whatsapp/ voice.

- **Treatment Recommendations**

  Provides chemical (dosage) and organic remedies immediately after detection .

- **Value- Add**

  Integrates Real- Time Market Data (Agmarknet) to ensure farmer earn profits to afford pest control.

Innovation & Uniqueness

- **Smartphone- First Strategy**

  While drone are expensive, our 'Camera - first' approach makes this accessible to most of marginal farmer immediately.

- **Holistic Loop**

  We don't just detect pests; we guide the farmer on where to buy the medicine and how to sell the saved crop (using Agmarknet data).

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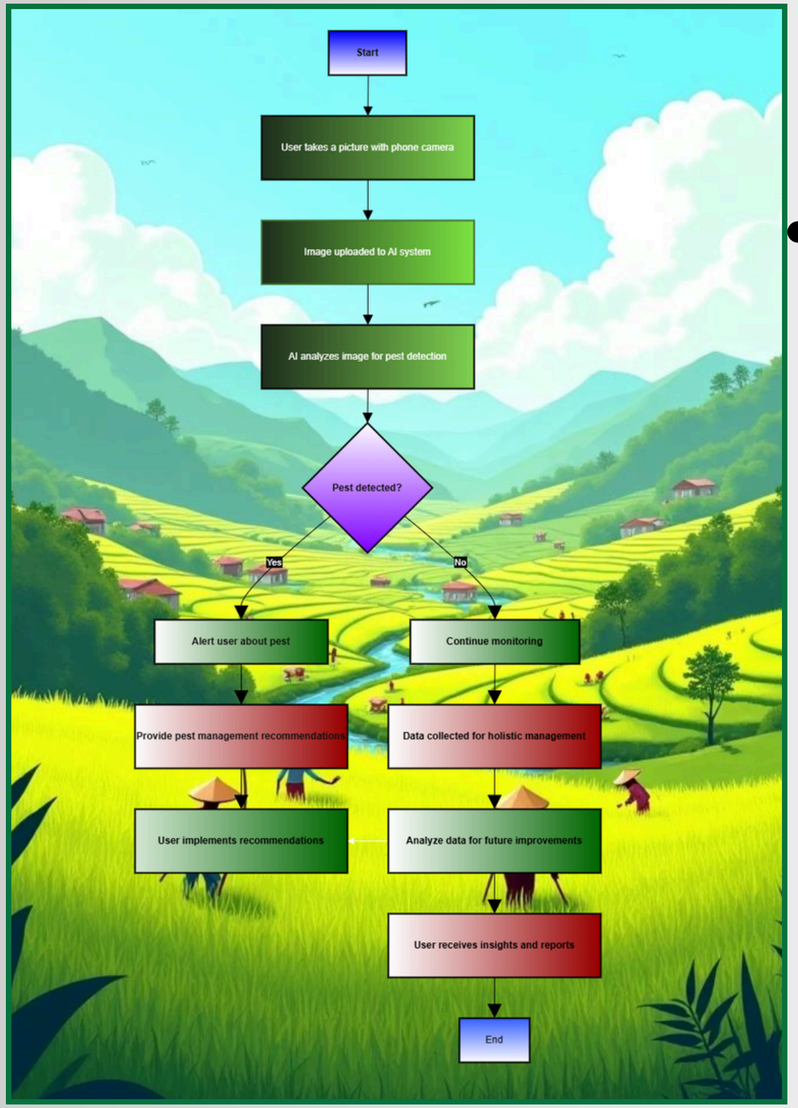
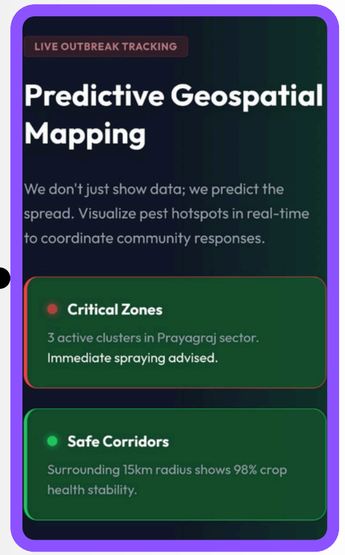
TECHNICAL APPROACH

TECHNOLOGY STACK

- **FRONTEND :** Html ,Css,Js (Web Based)
- **BACKEND :** Python (Flask)
- **API Services :** Weather , Google Map
  Text=to-Speech ,Agmarknet
  (Govt Data for Selling Crops)
- **DEPLOYMENT :** Render,Github Pages
- **DATABASE:** MongoDB

Methodology





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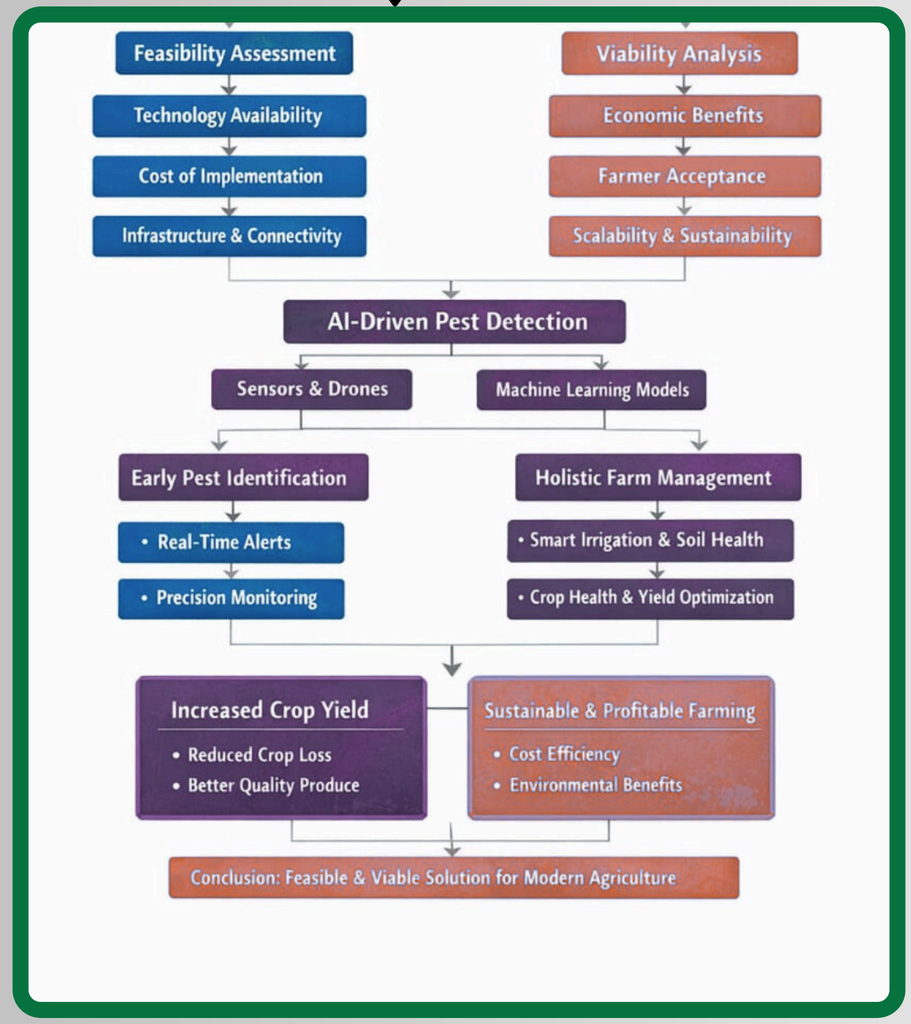
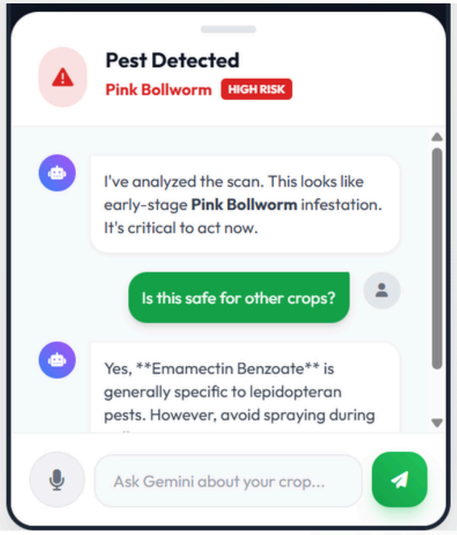
FEASIBILITY AND VIABILITY





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IMPACTS AND BENEFITS

POTENTIAL IMPACTS

1. AGRICULTURE & CROP IMPACT

- Early pest identification before visible damage occurs.
- Reduced crop losses and improved yield quality
- Timely interventions based on real-time field data.
- Better disease - pest differentiation

2. ENVIRONMENTAL IMPACTS

- Reduced chemical pesticide usage ,minimizing soil and water pollution.
- Preservation of beneficial insects.
- Lower carbon footprint .
- promotion of sustainable.

3. POLICY & NATIONAL IMPACTS

- Supports precision agriculture initiatives.
- Aligns with climate - smart agriculture goals.
- Enhance national food security.
- Supports data - backed agriculture policies.

BENEFITS

1. BENEFITS OF AI- DRIVEN EARLY PEST DETECTION

- Early identification of pest
(AI analyzes images from mobile phone,camera)
- Reduced crop losses.
- Lower Pesticides usage.
- Improved Accuracy.

2. BENEFITS OF HOLISTIC FARM MANAGEMENT

- Data- Driven Decision Making
(integrates pest data with weather, soil health)
- Optimized Resources Utilization .
- Cost Savings.
- Higher crop productivity.

3. LONG - TERM BENEFITS

- Improved Farm Planning
- Better Market Readliness

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RESEARCH AND REFERENCE

REFERENCES & RESEARCH

- **NITI Aayog**

  "NITI Aayog act as the policy driver, innovation enabler,and scaling authority that integrates AI- Driven early pest detection into India's holistic farm management ecosystem
  https://niti.gov.in/sites/default/files/2025-10/Reimagining_Agriculture_Roadmap_for_Frontier_Technology_Led_Transformation.pdf

- **ICAR**

  ICAR acts as the scientific backbone of AI-Driven early pest detection and holistic farm management by providing data , validation, regional expertise, and farmer outreach for sustainable & precise agriculture.
  https://www.icar.org.in/sites/defsult/files/2025-11/December-2025%20Indian%20Farming.pdf

- **Indian Agriculture Challenges2025**

  It is use to promoting innovation,enabling real-time monitoring , supporting sustainable farming ,& enssuring scalable, farmer-centric solutions across India.
  https://farmonaut.com/asia/indian-agriculture-challenges-2025-top-issues-solutions

- KISAAN SAHAYAK COMPETITIVE ANALYSIS

Feature	AGRIPLUS (Our Solution)	Plantix (Global)	DeHaat (Start-up)
Disease Detection	✓ AI+ Offline mode	✓ Good	✗ Manual
Land Measurement	✓ Self service (Satellite)	✗ No	✗ Field Agent Only
Market Linkage	✓ Live Price Compass(Govt API) API)	✗ No	✓ Supply chain
Local Community	✓ Hyper local chat	✓ Global forum	✗ None

Demo Link : https://themukeshdev.github.io/AgriPlus/

THANKYOU SO MUCH FOR GIVING YOUR TIME !

TEAM

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